

Solution_ Data Interpretation Part-1

Answer :

I. 1. 4

2. 2

3. 5; Statistics=135

4. 2; Marks obtained by Shweta in Sc. = 60

Marks obtained by Jyoti in S.St. = 80

$$\text{Required \%} = \frac{60}{80} \times 100 = 75\%$$

5. 1; Average marks obtained by all students in Hindi =

$$= \frac{(30+42+50+36+42)}{5} = \frac{200}{5} = 40$$

II. 1. 4; Marks obtained by Alka in Maths = 60% of 75 = 45

Marks obtained by Kushalin Economics = 50

Reqd. Ratio = 45 : 50 = 9 : 10

2. 2

3. 2; Total marks obtained by Prahlad in all the subjects =

$$= (60\% \text{ of } 125 + 88\% \text{ of } 100 + 82\% \text{ of } 75 + 66\% \text{ of } 150 + 30\% \text{ of } 125) = 75 + 88 + 61.5 + 99 + 37.5 = 361$$

Total maximum marks in all subjects = 575

$$\text{Required \%} = \frac{361}{575} \times 100 = 62.78$$

4. 3; need not to arrive at marks obtained, we can find the answer with % of marks which is given in the table.

5. 5; Total Marks obtained by shalin = 50 + 85 + 37.50 + 72 + 75 = 319.5
Average marks = 319.5 / 5 = 63.9

Total Marks obtained by Alka = 87.50 + 84 + 45 + 72 + 100 = 388.5

Average marks = 388.5 / 5 = 77.7

Reqd. Difference = 13.8

III. 1. 2

2. 1

3. 3; C.P of 1000 shares = Rs. 75,000

S.P of 500 shares each on Day 6 & 7 = (64 x 500) + (68 x 500) = 32,000 + 34,000 = Rs. 66,000

Total loss = 75,000 – 66,000 = 9,000

4. 5; Difference between the S.P of one share = 588 – 567 = 21

Total difference = 21 X 100 = Rs.2,100

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5. 1; Cost per stock on Day 1 = Rs.52

Cost per stock on Day 5 = Rs.62

$$\text{Required \%} = \frac{10}{52} \times 100 = 19.23\%$$

IV. 1. 3; Pop. of City D in 1990 = 130% of 135% of 8% of 3cr = 42,12,000

$$2. 2; \text{Reqd. \%} = \frac{\text{Population of B in 1990}}{\text{Population of B in 1970}} \times 100 =$$

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$$= \frac{120\% \text{ of } 145\% \text{ of } 17\% \text{ of } 3\text{cr.}}{17\% \text{ of } 3\text{cr.}} \times 100 = \frac{120 \times 145}{100 \times 100} \times 100 = 174\%$$

3. 4; Reqd. Ratio = 125% of 21% of 3 cr. :
120% of 17% of 3 cr. =
= 125 x 21 : 120 x 17 = 175 : 136

4. 2; Since % increase is asked, assume that population in 1970 was 100.

∴ Population in 1990 = 125% of 135% of 100 = 168.75

∴ % increase = 68.75%

5. 5; 31% of 3 cr = 93,00,000

V. 1. 2; Reqd. % = $\frac{12\% \text{ of } 410}{12.5\% \text{ of } 500} \times 100 = 78.72$
or 79%

2. 1; expenditure of Company B in the year 2013 = 540

Reqd. Average expenditure of Company B over the years = $\frac{510+300+420+300+680+540}{6}$
= $\frac{2750}{6} = 458$ ⇒ Difference = 540 – 458 = 82

3. 4; Income of Comp. A in the year 2009 = 114 % of 580 = 661.2

Income of Company C in the year 2009 = 125 % of 480 = 600

Required Difference = 661.2 – 600 = 61.2

4. 2; Expenditure of Company A in the year 2011= 320

Expenditure of Company B in the year 2011= 300

$$\text{Reqd. \%} = \frac{(320-300)}{320} \times 100 = \frac{20}{320} \times 100 = 6.25$$

or 6%

5. 3; Profit earned by Comp. A in 2008 = 12% of 410 = 49.2 or 49

Profit earned by Company B in 2008 = 10.5% of 510 = 53.55 or 54

Profit earned by Company C in 2008 = 12.5% of 500 = 62.5 or 63

Reqd. Ratio = 49 : 54 : 63

VI. 1. 3; Percent profit earned by Company Z in 2012= 45%

∴ 45% of Expenditure = 666

$$\Rightarrow \text{Expenditure} = \frac{666 \times 100}{45} = 1,480$$

∴ Total income = 1,480 + 666 = 2,146

∴ Reqd. Ratio = 2146 : 1480 = 29 : 20

2. 3; Percent profit earned by Company X in 2009 = 50%

∴ Profit = 50 % of 1,720 = Rs. 860 Lakh

Total income = 1,720 + 860 = 2,580

3. 2; Reqd. ratio = 65 : 5 : 50 = 13 : 1 : 10

4. 1; Percent profit earned by Company X in 2008 = 20%

∴ 20% of Expenditure = 312 ⇒

$$\text{Expenditure} = \frac{312 \times 100}{20} = 1,560$$

Percent profit earned by Company X in 2010 = 15%

∴ 15% of Expenditure = 225

$$\Rightarrow \text{Expenditure} = \frac{225 \times 100}{15} = 1,500$$

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Total Expenditure = 1,560 + 1,500 = 3,060

5. 4; Since, we don't have the data regarding the expenditure or income of the companies, we cannot find out the difference in expenditure.

VII. 1. 2; Average cost per tyre in the week

$$= \frac{2500 + 6000 + 5500 + 4500 + 1500 + 1800}{6} = \frac{21800}{6} = 3633.33$$

2. 3

3. 2; Cost per tyre manufacture on Thursday = Rs. 4,500

Cost per tyre manufactured on Tuesday = Rs. 6,000

$$\text{Reqd. \%} = \frac{4500}{6000} \times 100 = 75\% \quad 4. 1 \quad 5. 2$$

VIII. 1. 4; Total population of State C above poverty line = 75% of 620 = 465

$$\text{Total male population} = \frac{5}{6} \times 465 = 387.5$$

2. 1; Let us assume that total population below poverty line in State A = x $\therefore \frac{1}{4}$
 $x \times x = 10 \Rightarrow x = 10 \times 4 = 40$

3. 1; The female population above poverty line for State D = 336

$\therefore$ Male population above poverty line for State D = $\frac{1}{2} \times 336 = 168$

4. 2; Total population of State B = 550

Total population below poverty line in State B = 20% of 550 = 110

Total female pop. below poverty line in State B = $\frac{1}{2} \times 110 = 55$

Total population above poverty line in State B = 80% of 550 = 440

Total male population above poverty line in State B = $\frac{1}{2} \times 440 = 220$

$$\text{Reqd. \%} = \frac{55}{220} \times 100 = 25\%$$

5. 2; Total pop. above poverty line in State E = 70% of 930 = 651

IX. 1. 3; $\text{Reqd. \%} = \frac{50}{325} \times 100 = 15.38\%$

2. 5;

Branch	A	B	C	D	E
No.	325	350	450	325	275

3. 3; No. of officers who joined in 2013 = 500 $\Rightarrow 10\%$ of 500 = 50

4. 4; $\text{Reqd. Ratio} = 450 : 275 = 18 : 11$

5. 4

X. 1. 3; Average =

$$\frac{10000 + 12000 + 8000 + 10500 + 13000}{5} = 10,700$$

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2. 2; Cost Price of machine = Rs. 52,000 &
Profit = 15%

$$\text{Selling Price} = 52000 \times \frac{115}{100} = \text{Rs. } 59,800$$

3. 1; No. of machine sold by Company A =
 $10000 \times \frac{10}{100} = 1000$

Profit of Company A per machine = 50000
 $\times \frac{20}{100} = \text{Rs. } 10,000/-$

Total profit of Company A = $1,000 \times$
 $10,000 = \text{Rs. } 100 \text{ Lakh}$

No. of machine sold by Company C = 8000
 $\times \frac{35}{100} = 2,800$

Profit of Comp. C per machine = $52000 \times$
 $\frac{15}{100} = \text{Rs. } 7,800/-$

Total profit of Company C = $7,800 \times 2,800$
 $= \text{Rs. } 218.40 \text{ lakh}$

Difference between the net profit of
Company A & C = $218.40 - 100 = \text{Rs. } 118.40$
lakh

4. 1; machines sold by Company D =
 $10500 \times \frac{15}{100} = 1575$

machines sold by E = $13000 \times \frac{50}{100} = 6500$

$$\Rightarrow \text{Ratio} = \frac{1575}{6500} = 63:260$$

5. 1; No. of machines sold = 15% of 10,500
 $= 1,575$

Profit earned per machine = 40% of
 $35,000 = 14,000$

Total Profit earned = $1,575 \times 14,000 = \text{Rs.}$
 220.50 lakh